Lab: Skill Learning Analysis -- How You Get Better Without Trying

Objective

Investigate how **learning** works as a process of updating internal models through prediction errors, using real skills you are developing right now.

Prerequisites

- Completed modules on Systems through Action
- Read the module on Bayesian Updating

Part 1: Your Learning Curve

Goal: Track the learning process for a skill you are currently developing.

Choose a skill you have been practicing recently (a sport, an instrument, a video game, cooking, drawing). Answer:

1. **When you first started**: What did your internal model look like? What were your predictions about how to do it?
2. **Early mistakes**: What prediction errors did you experience? (Things that did not go as expected.)
3. **Current level**: How has your model updated? What do you now predict correctly that you used to get wrong?
4. **Remaining gaps**: Where do prediction errors still occur?

Write your response here

Part 2: The Prior Strength Experiment

Goal: Explore how the strength of your prior beliefs affects how quickly you learn.

Consider two scenarios:

Scenario A: A friend tells you a "fact" that contradicts something you are very confident about (e.g., "The capital of France is Berlin").

Scenario B: A friend tells you a "fact" about something you know nothing about (e.g., "The capital of Burkina Faso is Ouagadougou").

Feature	Scenario A (Strong Prior)	Scenario B (Weak Prior)
How confident were you before?		
Did you update your belief?		
How much evidence would you need to change your mind?		
In Active Inference terms, what was the precision of your prior?		

Write your response here

Part 3: Teaching as a Test of Learning

Goal: Test how well you have learned something by trying to teach it.

Choose one concept from this course so far (Markov blanket, prediction error, generative model, etc.). Explain it to someone who has never heard of it -- a friend, sibling, or family member.

1. What concept did you choose?
2. What analogy or example did you use?
3. What questions did they ask?
4. Where did you get stuck? (Those are the places where your own model still has gaps.)

Write your response here

Part 4: Prediction Error Diary -- Learning Edition

Goal: Track learning-specific prediction errors over three days.

Each day, record one moment where you learned something new because your prediction was wrong.

Day	What You Predicted	What Happened	What Your Brain Updated
1			
2			
3			

Write your response here

Discussion Questions

1. Why does making mistakes feel bad, even though prediction errors are essential for learning?
2. Is it possible to learn without prediction errors? Why or why not?
3. How does the idea of "precision" explain why some people are stubborn about changing their minds?

Write your response here

Summary Table

Concept	Definition	Your Example
Learning	Updating the parameters of a generative model over time	
Prior	The belief held before new evidence arrives	
Likelihood	How well the new evidence fits different possible explanations	
Posterior	The updated belief after incorporating new evidence	
Precision	The confidence assigned to a belief or sensory signal	